Runqi Pan (Roger)

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Research Focus: Brain-state-dependent neural responses and state-aware brain–computer interfaces

EDUCATION

Imperial College London *London, UK*

MEng Biomedical Engineering *Sep. 2023 – Jun. 2027 (Expected)*

- Current average: 74/100 (First Class level).
- Core Modules: Computational Neuroscience, Brain–Machine Interfaces, Adaptive Signal Processing and Machine Intelligence, Optimisation, Reinforcement Learning, Image Processing, Modelling in Biology.
- Master thesis 2026–27: *Does pre-stimulus brain state predict cortico-cortical evoked responses?*, supervised by Prof. Emmanuel M. Drakakis.

AWARDS

- **UTSIP Research Fellowship**, The University of Tokyo (2025).

PUBLICATIONS

- **Biomaterials**: Yu, X.*, Zeng, L., ..., **Pan, R.** (7th of 12), et al. "Blood-testis barrier-crossing extracellular vesicles for asthenozoospermia therapy via synergistic ATP replenishment and ferroptosis suppression." *Biomaterials* 327, 123777 (2026). doi:10.1016/j.biomaterials.2025.123777

Preprints and manuscripts

- **Pan, R.** "State or continuation? A paired no-stimulation pseudo-event control for pre-stimulus prediction of single-pulse evoked responses." TechRxiv preprint (planned, 2026); full paper in preparation for IEEE EMBC 2027.
- **Pan, R.**, Niinuma, T., Huang, Y., Sugino, M., Shimba, K., Kotani, K. "Target selection with a disappearing-stimulus mixed-reality P300 interface: feasibility and object-distinguishable versus uniform cues." arXiv:2611.13906 [cs.LG] (2026). With the Kotani laboratory.
- Chen, H., ..., **Pan, R.**, et al. "KalHuman: A frequency-aware Kalman framework for two-human close interaction pose estimation." Manuscript in preparation for *Neurocomputing*.
- Chen, H., ..., **Pan, R.**, et al. "TRELLIS-GPS: SMPL-guided, geometry-conditioned projection-aware sparse-latent diffusion for 3D human generation." Conference paper in preparation.

RESEARCH EXPERIENCE

Pre-stimulus Brain State and Evoked Responses, Imperial College London *London, UK*

MEng Individual Project | Prof. Emmanuel M. Drakakis, Department of Bioengineering *Oct. 2026 – Present*

- Built a causal (past-only) **MATLAB** pipeline testing whether pre-stimulus intracranial activity predicts cortico-cortical evoked potentials in open single-pulse electrical stimulation data (OpenNeuro ds004080), using nested model comparisons on held-out pulses, including a **Kalman state-space** model.
- Applied a paired no-stimulation **pseudo-event control** (following prior EEG and stimulation studies) in a nested out-of-sample comparison, with trial-level artefact covariates, to separate genuine state-dependence from continuing ongoing activity.
- Preregistered the confirmatory analysis on **OSF** (osf.io/c4fdw) before unlocking 50 held-out patients; in the confirmatory set (44 patients pooled), state features gave no useful gain beyond continuation.
- Designing an **analog-circuit phantom** with a planted effect, simulated in SPICE, to verify detection sensitivity.

Mixed-Reality P300 BCI for Assistive Wheelchair Navigation, University of Tokyo *Tokyo, Japan*

UTSIP Research Fellow | Prof. Kiyoshi Kotani, Graduate School of Frontier Sciences *Jun. 2025 – Aug. 2025*

- Designed and conducted a within-subject mixed-reality P300 study (6 participants, 2 cue conditions, 25 targets, 48 trials) with Unity/C#, HTC VIVE XR and three-channel (Fz/Cz/Pz) EEG at 500 Hz.
- Developed a **MATLAB** ERP pipeline (filtering, epoching, artefact rejection, feature extraction), retaining over 91% of epochs after signal-quality control.
- Implemented **shrinkage-LDA** decoding with nested trial-level validation: 18 of 48 held-out trials correct vs 1.9 expected by chance ($p < 10^{-12}$); linked decoded targets to Dijkstra path planning on an Arduino wheelchair.

Nanomedicine, Zhejiang University School of Medicine *Hangzhou, China*

Visiting Research Student | Prof. Xinghua Yu, Department of Urology & Andrology *Jun. 2024 – Sep. 2024*

- Contributed to an extracellular-vesicle nanoplatform (UCMSC-derived EVs loaded with ATP, NIR-II AIEgen-labelled) for fluorescence-guided delivery across the blood-testis barrier.
- Performed **computer-assisted sperm analysis** (12 kinematic parameters, 6 treatment groups) with one-way ANOVA and Tukey post-hoc tests, showing significant motility recovery in the EVs@ATP group ($p < 0.001$).

SKILLS & INTERESTS

- **Languages:** English (Fluent), Mandarin Chinese (Native)
- **Programming:** MATLAB, Python, C#
- **Neurotechnology & Data Analysis:** EEG acquisition, intracranial EEG / evoked-response analysis, ERP analysis, digital filtering, artefact rejection, LDA, state-space (Kalman) models, nested model comparison, bootstrap inference
- **Engineering Tools:** Unity, Arduino, Simulink, SPICE, Altium Designer, SolidWorks, Git, LaTeX